AKINYEMI FIYINFOLUWA

Electrical & Electronics Engineering Undergraduate | Embedded Systems, Automotive & IoT

📍 Lagos, Nigeria 📞 +234 9042764461 ✉️ inventorfiyin.2022@gmail.com 🔗 GitHub: inventorfiyin

🔗 LinkedIn: akinyemi-fiyinfoluwa

PROFESSIONAL SUMMARY

Results-oriented **Electrical & Electronics Engineering undergraduate** with focused technical specialization in **Embedded Firmware, Automotive Electronics, and IoT Telemetry Systems**. Proven track record deploying commercial-scale remote telemetry units across multi-phase industrial sites, designing 2-layer PCB hardware with power-conditioning and safety isolation, and building kinematic simulation environments with automated data telemetry. Adept at bridging physical circuit design, real-time control algorithms, and automated diagnostics to deliver fault-tolerant engineering solutions.

POSTSECONDARY EDUCATION

Bachelor of Engineering (B.Eng.) in Electrical and Electronics Engineering — **2023 – Expected 2028**

Covenant University, Ota, Ogun State, Nigeria

- **Relevant Coursework:** Embedded Computing, Microprocessor Systems, Control Systems Engineering, Signals & Systems, Power Electronics, Circuit Theory, Digital Electronics, Engineering Mathematics.

RELEVANT TECHNICAL EXPERIENCE

IoT Engineering Intern — *Uno Telos Nigeria* — **Jul 2026 – Sep 2026**

- Commissioned, configured, and diagnosed IoT-based telemetry devices across **12+ commercial client sites**, preserving uninterrupted data logging on multi-phase industrial power infrastructure.
- Executed AC/DC distribution checks, ground loop troubleshooting, and sensor signal integrity verification across **20+ telemetry nodes**, decreasing operational field faults by **30%**.
- Engineered multi-threaded Python automation scripts to audit and parse IP reputation logs across **1,700+ network endpoints**, decreasing diagnostic turnaround time by **95%** (from 5 hours to under 15 seconds).

Electrical Systems Lead & Operator — *Covenant University Chapel* — **Sep 2025 – Present**

- Supervise high-voltage stage power distribution, phase balancing, and console automation using the Avolites Titan platform during **10,000+ attendee** productions and live international broadcasts.
- Lead an **80+ member technical crew**, instituting structured pre-service load-testing protocols and isolated DMX control cabling to maintain zero power or signal disruptions.

Data Operations Coordinator — *Technical Crew Service Unit* — **Sep 2024 – Jul 2025**

- Automated operational scheduling, credential tracking, and technical rostering for an active **200+ member** production department.
- Standardized maintenance logs and stage asset allocation databases, reducing equipment scheduling conflicts by **40%** and speeding shift handovers.

SELECTED ENGINEERING PROJECTS

Autonomous Vehicle Kinematics & Battery SoC Simulator — **Python, NumPy, SciPy, Matplotlib**

- Engineered an autonomous vehicle modeling engine incorporating non-holonomic bicycle kinematics, global A* grid search, and Stanley path-tracking controllers for smooth trajectory tracking.
- Integrated an electrochemical multi-cell Li-ion battery pack discharge model with Coulomb-counting State-of-Charge (SoC) estimation, accounting for internal resistance (R_{int}) voltage drops across varying C-rate loads.
- Automated 2D tracking animation and cell discharge curve diagnostics, verifying trajectory accuracy and low-voltage protection thresholds within a **2% margin of error** against experimental data.

CU-PowerSentry: Smart Energy Telemetry & Protection System — **ESP32, KiCad, AC Telemetry, Embedded C**

- Developed a smart energy telemetry unit sampling 230V mains AC with non-invasive current (SCT-013) and voltage (ZMPT101B) transducers coupled to an ESP32 microcontroller.
- Drafted a custom 2-layer PCB in KiCad featuring integrated AC-DC switching regulation (HLK-PM01), optocoupler relay isolation, MOV surge suppression, and dedicated RF keep-out zones.
- Implemented real-time RMS active power computation and autonomous microsecond trip logic to safeguard connected loads against surges, under-voltage, and short-circuit faults.

Industrial Fuel Consumption Telemetry & Gateway Assembly — **Industrial IoT, Field Deployment, FreeCAD**

- Engineered and deployed remote telemetry units across **5+ commercial diesel generator facilities** to support continuous fuel-level monitoring and operating-hour tracking.
- Integrated submersible level transducers, industrial relays, and cellular IoT gateways into custom DIN-rail enclosures built to withstand high vibration and harsh thermal environments.

TECHNICAL & PROFESSIONAL COMPETENCIES

Embedded & IoT:	ESP32, Arduino, Peripheral Interfacing (I2C, SPI, UART, ADC), Sensor Telemetry, Power Supply Conditioning, Relay Logic.
Automotive & Controls:	CAN Bus (ISO 11898), OBD-II Diagnostics, SavvyCAN, Vector CANoe Fundamentals, BMS Telemetry, DC/BLDC Motor Control, PWM Regulators.
Hardware & PCB Design:	KiCad (Schematic Capture & PCB Layout), 2-Layer Routing, Prototyping, Breadboarding, PCB Soldering, Signal Verification, FreeCAD.
Programming & Tools:	Python (NumPy, SciPy, Matplotlib, Asyncio, Sockets), C, C++, Embedded C, MATLAB, Git, GitHub, VS Code, Linux CLI.
Leadership & Operations:	Cross-functional Team Leadership, Technical Documentation, Event Electrical Safety, Preventive Maintenance, Load Auditing.

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Head Football Coach — *Covenant University Athletics* — **Sep 2023 – Present**

- Honored as **Coach of the Tournament** after strategizing and leading the collegiate squad to a tournament championship victory.
- Direct physical conditioning, tactical match execution, and weekly performance reviews for a 22-player squad, cultivating discipline, rapid decision-making, and high team morale.

REFERENCES

Available upon request.